COMSATS UNIVERSITY ISLAMABAD

Final Project Proposal

Disaster Relief Management System

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1. Abstract

Natural disasters often cause severe human and infrastructural losses, and efficient coordination of relief operations becomes critical. Currently, there is a lack of integrated systems to connect victims, volunteers, and NGOs in real time. Existing manual or isolated solutions delay resource delivery and hinder data sharing between stakeholders. The proposed Disaster Relief Management System (DRMS) aims to address these issues by creating a centralized web platform for disaster response coordination. It bridges the gap between relief providers and recipients by enabling transparent communication, tracking resource distribution, and maintaining real-time situational awareness. The main objectives include improving coordination, reducing response time, and ensuring equitable allocation of relief materials. The DRMS will provide a reliable and data-driven interface for efficient disaster management and resource distribution.

2. Introduction

The purpose of this document is to present the vision, objectives, and scope of the Disaster Relief Management System (DRMS). This project is designed to assist government agencies, NGOs, and volunteers in efficiently coordinating relief efforts during natural and human-made disasters. In the current environment, disaster management activities are often managed manually or through fragmented tools, leading to miscommunication, resource misallocation, and delays in critical decision-making.

The proposed DRMS aims to create an integrated and centralized platform that enables stakeholders to register, communicate, and track aid distribution in real time. This system also provides an opportunity to digitally record disaster data, analyze needs, and monitor progress effectively. It will serve as a web-based interface accessible to NGOs, donors, victims, and administrative authorities. The document further defines the problem, objectives, modules, and the tools used to build the system.

3. Problem Statement

During disaster situations such as floods, earthquakes, or wildfires, the lack of a coordinated relief system leads to inefficiencies and delays in aid distribution. Relief organizations often operate independently, causing duplication of efforts and leaving certain areas underserved. Victims struggle to communicate their needs to the authorities or locate nearby assistance centers. Existing systems are mostly manual, relying on phone calls, paperwork, or informal communication, which becomes unreliable under crisis conditions.

There is also no real-time database that tracks resources, volunteers, and aid distribution statistics. The absence of a centralized digital platform results in delayed responses and wastage of valuable resources. The proposed Disaster Relief Management System will resolve these challenges by automating the entire relief coordination process using an online web-based platform connected to a structured SQL database.

4. Problem solution/objective of proposed system

During disasters, effective communication and resource coordination are critical to saving lives. However, most existing relief efforts rely on manual communication, fragmented data sources, and uncoordinated workflows between NGOs, volunteers, and victims. The proposed Disaster Relief Management System (DRMS) aims to provide a centralized and automated web platform that bridges this gap. This system will allow disaster victims to register their needs online, NGOs to track and allocate resources, and volunteers to receive assignments efficiently. By digitizing relief coordination, the DRMS will reduce delays in response and improve the accuracy of resource distribution.The system will maintain a real-time SQL database of available resources, victims' requests, and active volunteers. This will ensure transparency in operations and accountability among participating organizations. In addition, administrators can monitor disaster zones using interactive dashboards to make informed decisions. The DRMS will also integrate notification alerts to inform victims and volunteers about updates or resource deliveries. Through automation, the system minimizes redundant efforts among multiple relief agencies working in the same region. It also helps prevent resource wastage by tracking inventory levels and identifying underserved areas. Overall, the platform enhances collaboration and ensures equitable delivery of aid. The proposed system contributes to sustainable disaster management by leveraging digital technology to streamline critical operations. Ultimately, the DRMS supports faster, data-driven, and more coordinated disaster responses, thereby improving recovery efficiency and saving more lives.

Objectives:

BO-1: Reduce disaster relief response time by 30% through real-time data sharing and automated coordination.

BO-2: Improve transparency and accountability by digitally tracking 100% of donations and resource distributions.

BO-3: Minimize redundant relief efforts and resource wastage by at least 25% using centralized reporting and monitoring.

5. Advantages of proposed system

- ## Improved Coordination

  The Disaster Relief Management System provides a centralized platform that improves coordination between NGOs, volunteers, government agencies, and disaster victims.

- ## Faster response time

  Real-time communication and automated workflows help reduce delays in relief operations and ensure quicker delivery of aid to affected areas.

- ## Efficient Resource management

  The system tracks supplies, donations, and inventory effectively, reducing resource wastage and preventing duplication of efforts.

- ## Transparent operations

  Digital tracking of donations, volunteer activities, and aid distribution increases transparency and accountability among participating organizations.

- ## Real time data access

  Administrators and relief organizations can access updated disaster information, victim records, and camp details instantly through a centralized database.

- ## Scalability and accessibility

  Being a web-based platform, the system can be accessed from different locations and can be expanded in the future with additional features and technologies.

- ## Enhanced decision making

  Reports, analytics, and dashboards help administrators analyze disaster situations and make informed decisions quickly.

6. Vision Statement

For disaster relief organizations, volunteers, and affected individuals who need a fast, transparent, and coordinated way to manage aid, resources, and communication during disasters, the Disaster Relief Management System (DRMS) is a web-based and mobile-enabled platform. It facilitates real-time disaster response, resource tracking, volunteer coordination, and victim assistance through a centralized digital network. Unlike current manual and fragmented relief management methods that lead to delays, duplication, and inefficiency, our product provides an integrated and data-driven solution that ensures faster response times, equitable resource distribution, and improved collaboration among all stakeholders.

7. Scope

The Disaster Relief Management System (DRMS) is designed to provide a centralized platform for managing disaster response activities efficiently. The system will connect victims, volunteers, NGOs, and government agencies through a web-based interface. It will support victim registration, volunteer coordination, resource allocation, shelter management, and donation tracking. The system will maintain a real-time database for monitoring relief operations, managing supplies, and generating reports for administrators. It will also provide secure login access and automated notifications for users. The project scope is limited to disaster registration, victim management, volunteer management, and resource distribution modules. Advanced features such as live GPS tracking, weather forecasting, and AI-based analytics are considered outside the scope of the current phase.

8. Modules

8.1 Module 1:ReliefCamp & victim Management

- **FE-1:** Register and manage victims affected by disasters.
- **FE-2:** Assign victims to specific relief camps based on location and camp capacity.
- **FE-3:** Track relief aid delivery and provide updates regarding camp occupancy and victim status

8.2 Module 2: Volunteer Management

- **FE-1:** Register and assign volunteers to relief camps or disaster-affected zones.
- **FE-2:** Track volunteer availability, skills, assigned duties, and participation details.
- **FE-3:** Send emergency notifications and maintain records of volunteer activities during relief operations.

8.3 Module 3: Supply and Resource Management

- **FE-1:** Record and track supplies such as food, medicine, clothing, and emergency equipment delivered to relief camps.

- **FE-2:** Maintain MongoDB logs of resource distributions and inventory records for each relief camp.
- **FE-3:** Generate supply status reports and analyze resource utilization using MongoDB aggregation pipelines.

8.4 Module 4: Disaster Reporting and Analytics

- **FE-1:** Generate disaster summary reports and relief progress reports using SQL queries.
- **FE-2:** Create basic dashboards and data visualizations for monitoring relief activities and disaster statistics.
- **FE-3:** Support analysis of affected regions, victim statistics, and relief effectiveness for better decision-making.

8.5 Optional Module: GUI/Web Interface

- **FE-1:** Develop a simple HTML/PHP-based interface for data entry, monitoring, and report viewing.
- **FE-2:** Connect forms and interfaces to MongoDB collections including Victims, Volunteers, ReliefCamps, Supplies, and Disasters.
- **FE-3:** Provide basic CRUD operations (Create, Read, Update, Delete) for testing and system functionality verification.

9. System Limitations/Constraints

LI-1: Internet access is mandatory for real-time system operations and communication.

LI-2: System performance depends on the quality of the hosting server, network bandwidth, and database response time.

LI-3: Geolocation and location-tracking accuracy depend on user device permissions and GPS availability.

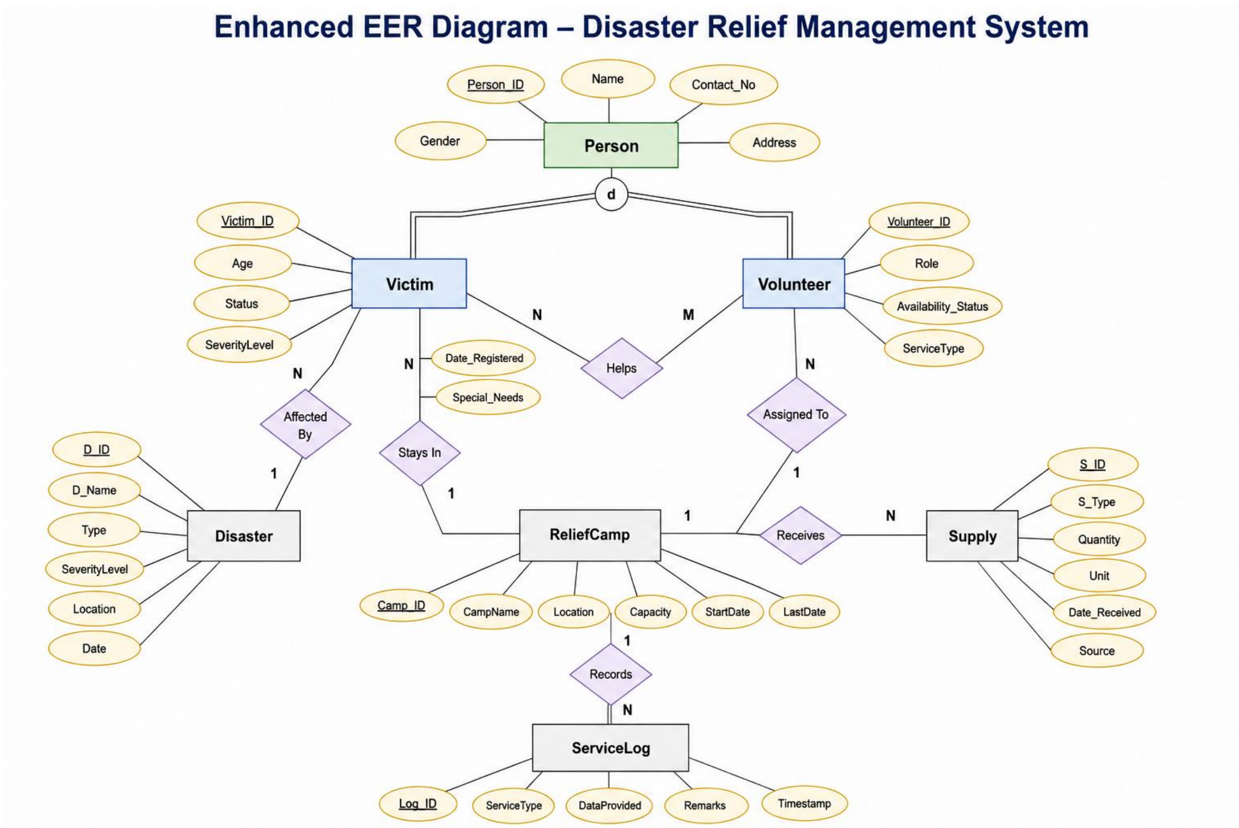
LI-4: Disaster-related data and reports must be verified and validated by local authorities to ensure accuracy and reliability.

10. Tools and Technologies

Tools	Versions	Rationale
Visual Studio Code	1.94	IDE
MongoDB Server	Latest	NoSQL Database
GitHub	Latest	Version Control
Oracle SQL	Latest	Rational Database
Python	3.10	Back-end Development

The relational database will also be converted into a MongoDB document-based schema using JSON collections and aggregation pipelines.

11. Entity relationship Diagram



The database design implements Enhanced Entity Relationship (EER) concepts including superclass-subclass relationships, disjoint specialization, total participation, partial participation, and many-to-many relationships. A superclass named “Person” is specialized into subclasses “Victim” and “Volunteer”.

12. DATABASE NORMALIZATION:

The relational database schema of the Disaster Relief Management System (DRMS) is normalized to reduce data redundancy and maintain data consistency. The normalization process is performed through First Normal Form (1NF), Second Normal Form (2NF), and Third Normal Form (3NF), and the final schema is refined up to Boyce-Codd Normal Form (BCNF).

This normalization process will ensure that all tables are properly structured with minimal duplication of data and efficient dependency management. It will also improve database integrity, optimize storage efficiency, and support reliable query execution during disaster management operations. The normalized schema will help maintain accurate relationships between entities such as Victims, Volunteers, Relief Camps, Supplies, and Disasters.

13. Queries

- ## Situation 1

  View all victims along with the disaster they were affected by and the relief camp they were assigned to.

```
> db.victims.aggregate([
  {
    $lookup: {
      from: "relief_camps",        // Join with Relief Camps
      localField: "Camp_ID",
      foreignField: "Camp_ID",
      as: "camp_info"
    }
  },
  { $unwind: "$camp_info" },       // Flatten the array (make it a single object)
  {
    $lookup: {
      from: "disasters",           // Join with Disasters
      localField: "camp_info.D_ID",
      foreignField: "D_ID",
      as: "disaster_info"
    }
  },
  { $unwind: "$disaster_info" },
  {
    $project: {                    // Select specific fields to show
      VictimName: "$Name",
```

```
      }
    },
    { $unwind: "$disaster_info" },
    {
      $project: {                    // Select specific fields to show
        VictimName: "$Name",
        CampName: "$camp_info.CampName",
        DisasterName: "$disaster_info.D_Name"
      }
    }
  ])
< {
    _id: ObjectId('69347db8c176ccaea40897a9'),
    VictimName: 'M Ali',
    CampName: 'Islamabad Relief Camp',
    DisasterName: 'Flood'
  }
```

- ## Situation 2

Find the total quantity of supplies delivered to each relief camp.

```
> db.victims.aggregate([
    {
      $lookup: {
        from: "relief_camps",
        localField: "Camp_ID",
        foreignField: "Camp_ID",
        as: "camp_data"
      }
    },
    { $unwind: "$camp_data" },
    {
      $group: {
        _id: "$camp_data.CampName",   // Group by Camp Name
        TotalVictims: { $sum: 1 }     // Count 1 for each victim found
      }
    },
    { $sort: { TotalVictims: -1 } }  // Sort descending
  ])
< {
    _id: 'Islamabad Relief Camp',
    TotalVictims: 1
  }
```

- ## Situation 3

Find victims who are in camps that have more than 100 people capacity.

```
> db.volunteers.aggregate([
    {
      $lookup: {
        from: "relief_camps",
        let: { camp_id: "$Camp_ID" },
        pipeline: [
          {
            $match: { $expr: { $eq: ["$Camp_ID", "$$camp_id"] } }
          },
          {
            $lookup: {
              from: "disasters",
              localField: "D_ID",
              foreignField: "D_ID",
              as: "disaster"
            }
          },
          {
            // ### UPDATED LINE BELOW ###
            $match: { "disaster.D_Name": "Flood" }
          }
        ],
```

```
            // ### UPDATED LINE BELOW ###
            $match: { "disaster.D_Name": "Flood" }
          }
        ],
        as: "matched_camp"
      }
    },
    { $match: { matched_camp: { $ne: [] } } },
    { $project: { Name: 1, Role: 1 } }
  ])
< {
    _id: ObjectId('69347fc3c176ccaea40897ab'),
    Name: 'Farhan Jaffar',
    Role: 'Medic'
  }
  {
    _id: ObjectId('693480bcc176ccaea40897ad'),
    Name: 'Ahmad',
    Role: 'Nurse'
  }
```

- **Situation 4**

  Find all people who are either volunteers or victims

```
> db.volunteers.aggregate([
    {
      $lookup: {
        from: "relief_camps",
        let: { camp_id: "$Camp_ID" }, // Define variable from Volunteer
        pipeline: [
          {
            $match: { $expr: { $eq: ["$Camp_ID", "$$camp_id"] } } // Match Camp ID
          },
          {
            $lookup: {
              from: "disasters",
              localField: "D_ID",
              foreignField: "D_ID",
              as: "disaster"
            }
          },
          {
            $match: { "disaster.D_Name": "Punjab Floods 2025" } // The "Subquery" logic
          }
        ],
```

```
        $match: { "disaster.D_Name": "Punjab Floods 2025" } // The "Subquery" logic
      }
    ],
    as: "matched_camp"
  }
},
{ $match: { matched_camp: { $ne: [] } } }, // Only keep volunteers where a match was fou
{ $project: { Name: 1, Role: 1 } }
])
<
> db.victims.aggregate([
  {
    $project: { Name: 1, Type: "Victim" } // Select Name, label as Victim
  },
  {
    $unionWith: {
      coll: "volunteers",                // Combine with Volunteers collection
      pipeline: [ { $project: { Name: 1, Type: "Volunteer" } } ]
    }
  },
  { $sort: { Name: 1 } }                 // Sort alphabetically
])
< {
```

```
< {
    _id: ObjectId('693480bcc176ccaea40897ad'),
    Name: 'Ahmad',
    Type: 'Volunteer'
  }
  {
    _id: ObjectId('69347fc3c176ccaea40897ab'),
    Name: 'Farhan Jaffar',
    Type: 'Volunteer'
  }
  {
    _id: ObjectId('69347db8c176ccaea40897a9'),
    Name: 'M Ali',
    Type: 'Victim'
  }
```

14. Conclusions

The Disaster Relief Management System provides a scalable and efficient platform for managing disaster response activities. By integrating SQL and MongoDB technologies, the system ensures effective coordination, transparency, and optimized resource distribution during emergencies.

15. References

[1] United Nations Office for Disaster Risk Reduction (UNDRR), "Global Assessment Report on Disaster Risk Reduction." Available: UNDRR Global Assessment Report. [Accessed: Oct. 10, 2025].

[2] A. Coppola, *Introduction to International Disaster Management*. Oxford: Butterworth-Heinemann, 2015, pp. 45–212.

[3] S. Ahmed and K. Iqbal, "Web-Based Disaster Management System Using Cloud Infrastructure," *International Journal of Computer Applications*, vol. 183, pp. 25–31, Jun. 2021.

[4] K. Nakamura, T. Tanaka, and M. Sato, "A Cloud-Based Real-Time Disaster Response Platform," in *Proceedings of the International Conference on Humanitarian Technology (HumTech)*, 2020, pp. 77–85.